AccessCast

One video. Five ways to watch.

[pending: e1.headline]

Gemma 4 on AMD Instinct MI300X · AMD Developer Hackathon ACT III · Create a New Kind of Experience

Most video says nothing to people who can't see it

- *[pending: cited.gbd.blind]* people are blind and *[pending: cited.gbd.mod_severe]* have moderate to severe vision loss.
- *[pending: cited.who.hearing]* people need rehabilitation for disabling hearing loss.
- *[pending: cited.oecd.literacy]* of adults read at or below the lowest literacy level.
- The largest free archive of described video holds about *[pending: cited.market.youdescribe_hours]* hours. YouTube received *[pending: cited.market.youtube_uploads]* hours a minute.

The law already asks for it

Where	Rule	When
EU	European Accessibility Act	*[pending: cited.law.eaa_date]*
Italy	D.Lgs. 82/2022, fines *[pending: cited.law.italy_fines]*	in force
India	MIB accessibility guidelines for OTT	*[pending: cited.law.india_mib]*
US	ADA Title II web rule (WCAG 2.1 AA)	*[pending: cited.law.ada_title2]*

One video, five experiences

"I can't see the screen" · "I can't hear" · "Simpler words, please" · "हिन्दी" · "Italiano"

All five are built from one analysis, so they never disagree. Profiles combine.

How it works

1. Listen: Whisper words, forced alignment, Gemma 4 E4B cross-checks the transcript and labels sounds.
2. Look: Gemma 4 31B lists what is visible, at temperature 0.
3. Plan: code finds the pauses and gives each line a word budget.
4. Write: Gemma writes only from the visible facts.
5. Check: code checks every line. Gemma repairs. A person approves.
6. Render: one MP4 per audience, loudness and timing checked.

The model proposes, code decides

- Rules tied to the real audio timeline: the line must fit the pause, measured in milliseconds.
- Names, claims, sounds, restated dialogue and injected instructions are blocked in code.
- Confidence is computed from measurements, never asked of the model.
- The server refuses to render while any line waits for a person.

Result: lines that fit the pauses

[pending: e01.table]

Test set: 10 W3C videos never used for tuning. Measured by a script that imports none of our checks.

More results

[pending: results.more]

On one AMD MI300X

[pending: amd.slide]

It holds up

[pending: reliability.slide]

Who pays, and what it costs

[pending: business.slide]

Our cost is GPU time plus a person's review time. List prices are labelled as list prices.

What didn't work, and what's next

[pending: negative.slide]

Prior work: AudioSight, ViddyScribe, AutoAD-Zero. Code: github.com/Prashant-thakur77/AccessCast · Live: accesscast.vercel.app